NLP Homework 3

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The source code of the project can be viewed on github: https://github.com/randnull/nlp

I. INTRODUCTION

This work aims to research an word2vec algorithms.

II. DATASET

The *ag news dataset* [1] was selected as main dataset for this work. It contains 127.500 samples of news titles and descriptions with 4 labels:

- World
- Sports
- Business
- Sci/Tech

III. WORD2VEC

Word2vec has two different types:

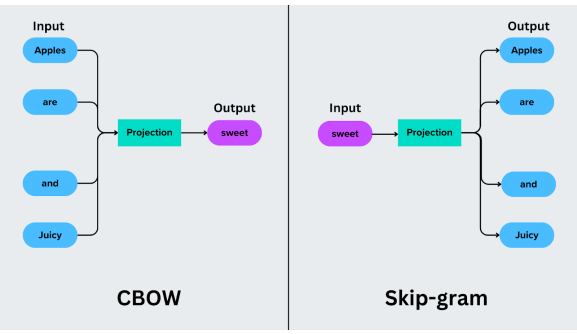

FIG. 1: skipgram vs cbow

The main difference lies in whether one is predicting the word based on the context (CBOW), or the context based on the word. (Skip-gram). In our experiment we will use skip-gram algorithm.

The next step is to do a PCA on our embeddings to see if they make sense.

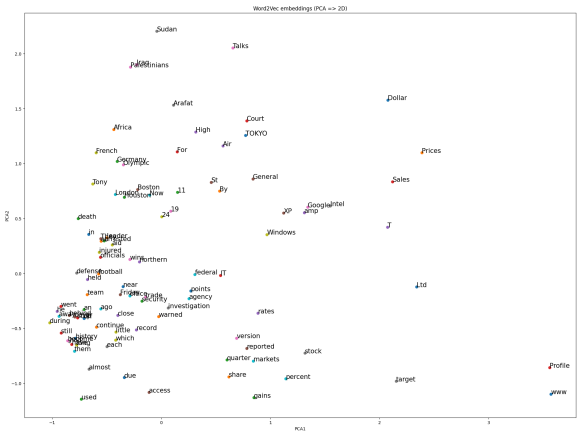

FIG. 2: Word2vec PCA

We see that similar points have formed their own clusters. For example, there is a group of companies - Google, Intel, XP. There is a group of languages - German, French. This shows that PCA found dependencies in the embeddings, that is, the embeddings are meaningful.

IV. INTRINSIC EVALUATION

We already check embedding for intrinsic evaluation in PCA example. Futhermore, we can get the closest words for a given using cosine similarity vectors.

```
w2v_model.most_similar("football", 10)
```

Word	Cosine Similarity
college	0.8105
coaching	0.7888
basketball	0.7746
coaches	0.7588
soccer	0.7581
vacancy	0.7346
Willingham	0.7334
hockey	0.7324
graduate	0.7272
varsity	0.7141

TABLE I: `w2v_model.most_similar("football", 10)`

These words appear in the same context as the word "football," which demonstrates the operation of cosine

similarity for vectors.

Other example aim is to show how positive/negative supports works:

```
w2v_model.wv.most_similar(positive=["university"],
negative=["school"])
```

Word	Similarity
department	0.4292
analyst	0.4028
store	0.3947
Federated	0.3820
Store	0.3781
operates	0.3770
TiVo	0.3747
Depot	0.3746
distributor	0.3743
Diebold	0.3729

TABLE II: `most_similar(positive=["university"], negative=["school"])`

The most likely word is "department," which makes sense, since a university involves both academic/government research and teaching, whereas a school involves only teaching.

V. EXTRINSIC EVALUATION

For extrinsic evaluation we will use our dataset for multiclasses classification task with logistic regression.

Due to we have a sentences for classification, but embeddings are avaliable only for word, we need to use methods to create embeddings from words.

There are two different types:

- Concatenate: create a long vector shape 1 x dN, with all embeddings
- Pool: take a mean from every embedding per position, presents in picture 3. We will use this method.

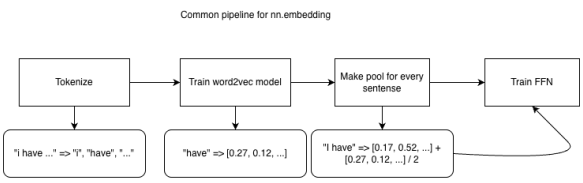

FIG. 3: common pipeline for w2v

After we have created our sentence embeddings, we will pass them to a logistic regression 4 with cross-entropy loss:

$$L = -\sum_{k=1}^{K} y_k \log(p_k)$$

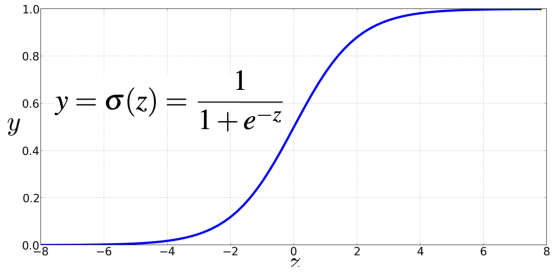

FIG. 4: Logistic regression

Now we can see our result metrics III:

	Precision	Recall	F1-score	Support
0	0.90	0.89	0.90	1900
1	0.94	0.96	0.95	1900
2	0.86	0.84	0.85	1900
3	0.85	0.86	0.86	1900
Accuracy			0.89	7600
Macro avg	0.89	0.89	0.89	7600
Weighted avg	0.89	0.89	0.89	7600

TABLE III: Classification report

As we can see, sufficiently high results have been achieved for each class.

VI. RESULT

As a result, we see that the word2vec model effectively represents word embeddings for both intrinsic and extrinsic contexts.

[1] Huggingface, *ag news*, https://huggingface.co/datasets/sh0416/ag_news (accessed: 22 March 2026).